

```

1. package mongodb;

import java.sql.Connection; import
java.sql.DriverManager; import
java.sql.ResultSet; import
java.sql.Statement;

public class Testing {

    public static void main(String[] args){

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/exercise",
                            "root","root123");

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("select * from students");    while(rs.next()) {

                System.out.println(rs.getInt(1)+" "+
rs.getString(2)+" "+rs.getInt(3));

            }

            con.close();

        }catch(Exception e){

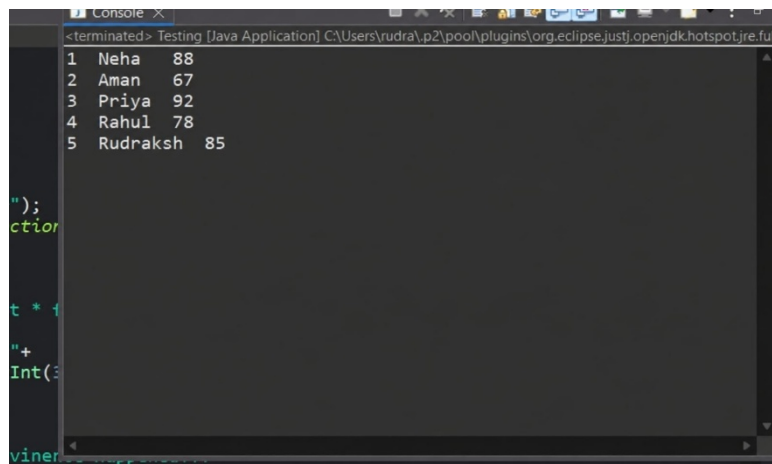
            System.out.println("Sorry for the inconvenience happened..! "
                            + "Check the inputs properly one more time");

        }

    }

}

```



The screenshot shows a Java application window titled "Console X" with the following output:

```
<terminated> Testing [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full\jre\bin\java.exe
1 Neha 88
2 Aman 67
3 Priya 92
4 Rahul 78
5 Rudraksh 85
```

Below the output, there is a snippet of Java code from the source editor:

```
");
ction
t * f
"+
Int(3
viner
```

```
2.CRUD_Test; package CRUD; import java.sql.Connection; import
java.sql.PreparedStatement; import java.sql.ResultSet;

public class CRUD_Test {

    public void create_data(String sl_no,String name,String email){

        DB_Connection obj_DB_Connection=new DB_Connection();

        Connection connection=obj_DB_Connection.get_connection(); PreparedStatement
ps=null;

        try {

            String query="insert into student(sl_no,name,email) values
(?,?,?)";

            ps=connection.prepareStatement(query);

            ps.setString(1, sl_no);

ps.setString(2, name);                ps.setString(3, email);

            System.out.println(ps);

ps.executeUpdate();

        } catch (Exception e) {

            System.out.println(e);

        }

    }

}
```

```

    }

    public void read_data(String sl_no){

        DB_Connection obj_DB_Connection=new DB_Connection();

        Connection connection=obj_DB_Connection.get_connection();

        PreparedStatement ps=null;

        ResultSet rs=null;

        try {

            String query="select * from student";

            ps=connection.prepareStatement(query);

            //ps.setString(1, sl_no);

            System.out.println(ps);

            rs=ps.executeQuery();

            while(rs.next()){

                System.out.println("Sl no -"+rs.getString("sl_no"));

                System.out.println("name -"+rs.getString("name"));

                System.out.println("email -"+rs.getString("email"));

                System.out.println("-----");

            }

        } catch (Exception e) {

            System.out.println(e);

        }

    }

}

    public void update_data(String sl_no,String new_sl_no,String name,String email){

        DB_Connection obj_DB_Connection=new DB_Connection();

        Connection connection=obj_DB_Connection.get_connection();

        PreparedStatement ps=null;

        try {

            String query="update student set sl_no=?,name=?,email=? where

```

```

sl_no=?";

        ps=connection.prepareStatement(query);
        ps.setString(1, new_sl_no);
ps.setString(2, name);                ps.setString(3, email);
        ps.setString(4, sl_no);
System.out.println(ps);                ps.executeUpdate();
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    public void delete_data(String sl_no){
        DB_Connection obj_DB_Connection=new DB_Connection();
        Connection connection=obj_DB_Connection.get_connection(); PreparedStatement
ps=null;
        try {
            String query="delete from student where sl_no=?";
            ps=connection.prepareStatement(query);
ps.setString(1, sl_no);                System.out.println(ps);
        ps.executeUpdate();
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    public static void main(String[] args) {
        CRUD_Test objTest=new CRUD_Test();

```

```

        //insert data  objTest.create_data("1", "jinu", "jinu@gmail.com");
objTest.create_data("2", "jawad", "jawad@gmail.com");

        //read data

        objTest.read_data("2");

        //update data          objTest.update_data("2", "22",
"jawad2", "jawad2@gmail");

        //delete data

        objTest.delete_data("1");

    }
}

```

2.1:

```

package CRUD; import
java.sql.Connection; import
java.sql.DriverManager;

```

```

public class DB_Connection {

    public Connection get_connection(){

Connection connection=null;

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            connection =
DriverManager.getConnection("jdbc:mysql://localhost:3306/exercise","root", "root123"); }
catch (Exception e) {

            System.out.println(e);

        }

        return connection;

    }
}

```

```

        public static void main(String[] args) {

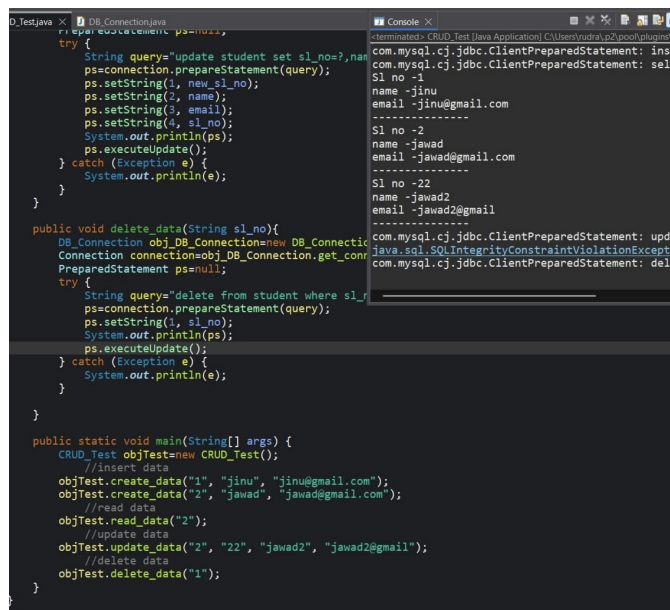
            DB_Connection obj_DB_Connection=new DB_Connection();

            System.out.println(obj_DB_Connection.get_connection());

        }
    }
}

```

OUTPUT:



The screenshot shows a Java IDE with two windows. The left window, titled 'D:\Testjava\...', displays the source code for 'DB_Connection.java'. The code includes methods for creating, reading, updating, and deleting data in a database. The right window, titled 'Console', shows the output of the program. It displays several SQL statements being executed, including 'insert', 'select', 'update', and 'delete', along with the resulting data rows. The output is formatted with headers and data rows separated by lines of dashes.

```

D:\Testjava\... DB_Connection.java
    try {
        String query="update student set sl_no=?,name=?,email=?";
        ps=connection.prepareStatement(query);
        ps.setString(1, new_sl_no);
        ps.setString(2, name);
        ps.setString(3, email);
        ps.setString(4, sl_no);
        System.out.println(ps);
        ps.executeUpdate();
    } catch (Exception e) {
        System.out.println(e);
    }
}

public void delete_data(String sl_no){
    DB_Connection obj_DB_Connection=new DB_Connection();
    Connection connection=obj_DB_Connection.get_connection();
    PreparedStatement ps=null;
    try {
        String query="delete from student where sl_no=?";
        ps=connection.prepareStatement(query);
        ps.setString(1, sl_no);
        System.out.println(ps);
        ps.executeUpdate();
    } catch (Exception e) {
        System.out.println(e);
    }
}

public static void main(String[] args) {
    CRUD_Test objTest=new CRUD_Test();
    //insert data
    objTest.create_data("1", "jinu", "jinu@gmail.com");
    objTest.create_data("2", "jawad", "jawad@gmail.com");
    //read data
    objTest.read_data("2");
    //update data
    objTest.update_data("2", "22", "jawad2", "jawad2@gmail");
    //delete data
    objTest.delete_data("1");
}
}

Console
<terminated> CRUD_Test [Java Application] C:\Users\rudra\...
com.mysql.cj.jdbc.ClientPreparedStatement: ins
com.mysql.cj.jdbc.ClientPreparedStatement: sel
sl no -1
name -jinu
email -jinu@gmail.com
-----
sl no -2
name -jawad
email -jawad@gmail.com
-----
sl no -22
name -jawad2
email -jawad2@gmail
-----
com.mysql.cj.jdbc.ClientPreparedStatement: upd
java.sql.SQLIntegrityConstraintViolationExcept
com.mysql.cj.jdbc.ClientPreparedStatement: del

```